Data Analytics and Mining

Errors in hypothesis testing, Chi-square tests

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Errors and Power

Two kinds of errors

A hypothesis test decides from **one sample**. Samples contain random error, so the decision can be wrong — even if we did everything correctly. There are exactly two ways to be wrong:

	Reality: H_0 is true (no effect)	**Reality: H_0 is false (effect exists)**
We reject H_0	**Type I error** — a false alarm. Probability α	Correct decision. Probability $1 - \beta$ = **power**
We fail to reject H_0	Correct decision. Probability $1 - \alpha$	**Type II error** — a missed effect. Probability β

We never know which column we are in: if we knew the truth, we would not need a test. So we cannot avoid errors; we can only **control how often** they happen.

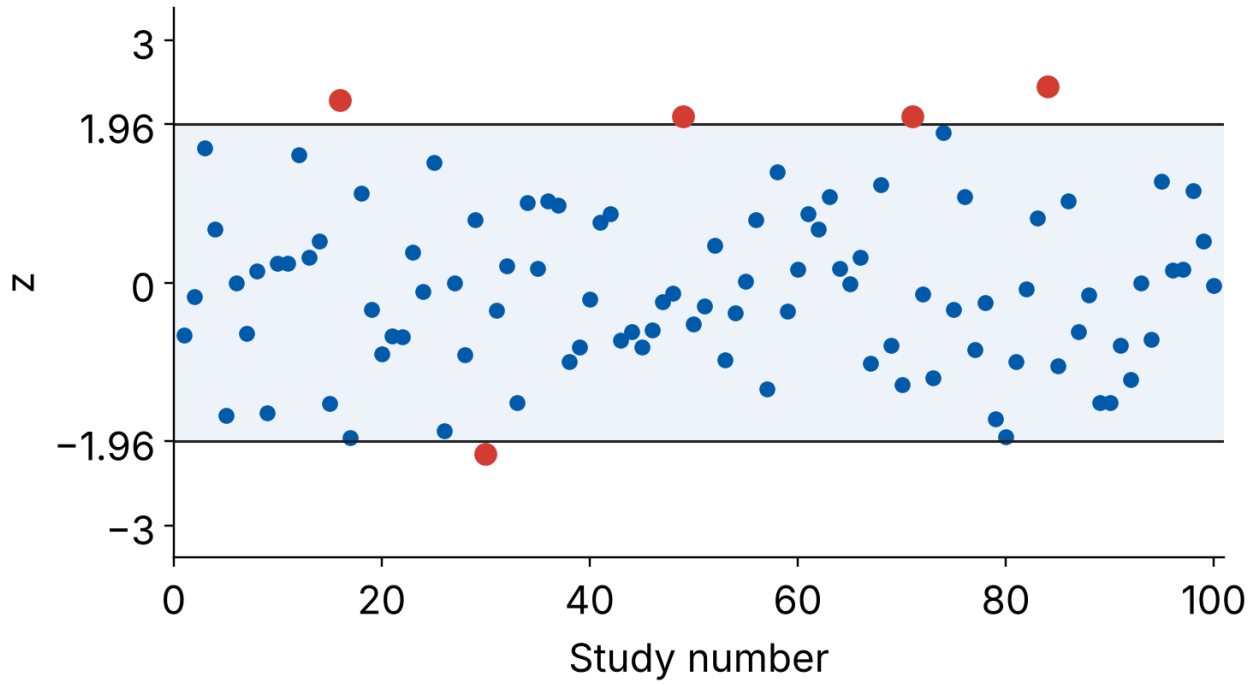
α and β are probabilities of the two errors. **Power** = $1 - \beta$ is the probability to find an effect that really exists.

Everyday examples

Situation	**H_0 ("nothing special")**	**Type I error (false alarm)**	**Type II error (missed effect)**
Court trial	The defendant is innocent	An innocent person is convicted	A guilty person goes free
Smoke alarm	There is no fire	The alarm rings, but there is no fire	There is a fire, but the alarm is silent
Medical test	The person is healthy	A healthy person is told "you are ill"	An ill person is told "you are healthy"
Our coffee study	The exam week changes nothing	We claim students drink more, but they do not	Students do drink more, but our test misses it

Which error is worse depends on the situation. Courts try hard to avoid Type I errors ("innocent until proven guilty"). A smoke alarm is tuned to avoid Type II errors: better a false alarm than a missed fire.

Type I error: false alarms



Imagine 100 studies of an effect that does **not** exist: H_0 is true in every study. Each study computes its own z.

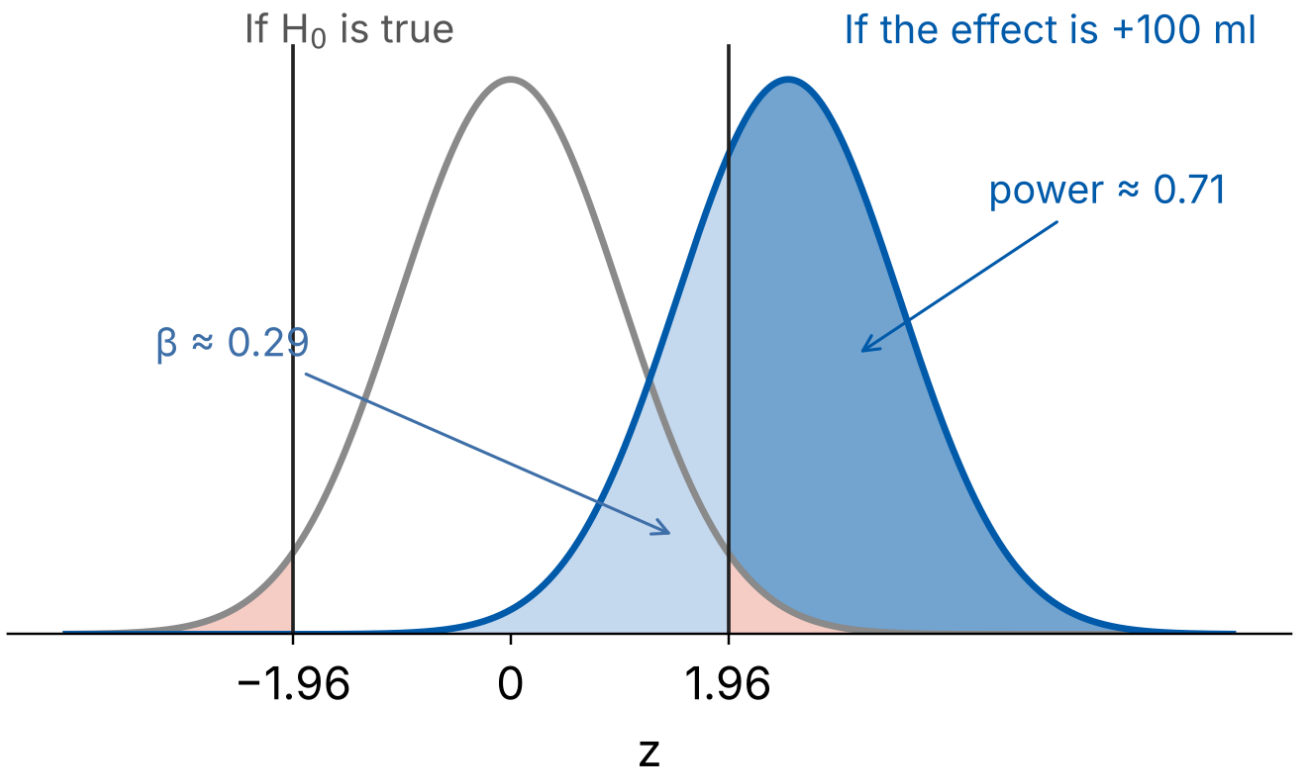
If H_0 is true, z follows the standard normal distribution, so about 5% of the z-values fall beyond ±1.96 just by chance.

Here 5 of the 100 studies (red points) would announce a "significant" effect that does not exist.

α is the probability of a Type I error. We choose it ourselves, before looking at the data: α = 0.05 means we accept 5% false alarms.

Caution: test 20 things that have no effect, and on average one of them will come out "significant" (20 × 0.05 = 1).

Type II error and power



Now suppose the effect is **real**: in the population the exam week adds +100 ml on average (σ = 400 ml, n = 100, so SE = 40 ml).

Then our z is centered not at 0 but at 100 / 40 = **2.5** (blue curve). We still use the same rule: reject H_0 if z is beyond ±1.96.

Power = probability to reject H_0 when the effect is real = the dark blue area ≈ **0.71**.

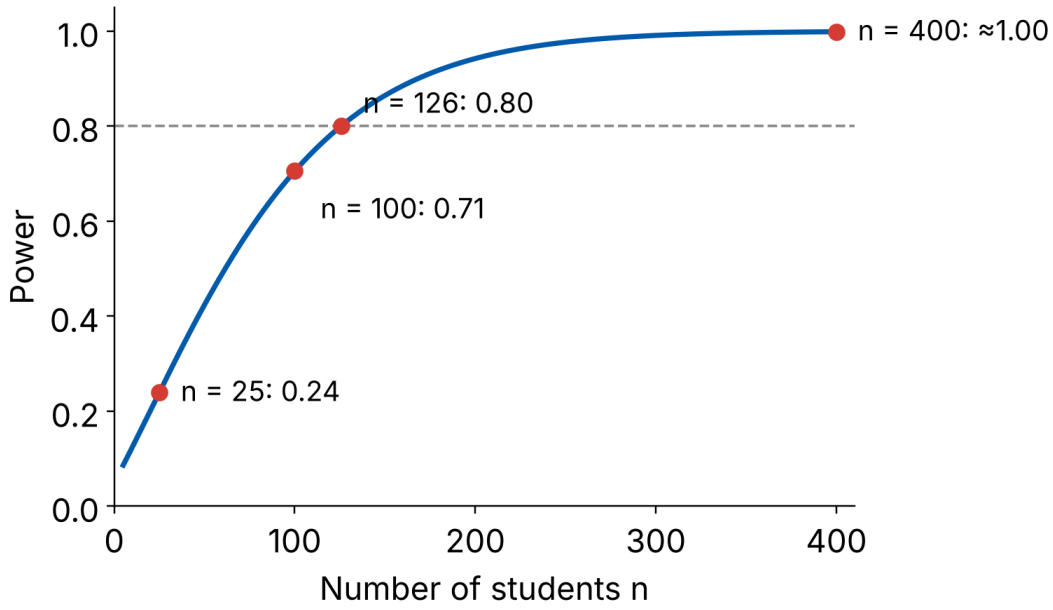
β = probability to miss the real effect = the light blue area ≈ **0.29**. Power = 1 − β.

Even with a real effect, about 3 out of 10 such studies would find "nothing".

How we got 0.71: z − 2.5 is standard normal, so P(z > 1.96) = P(standard normal > 1.96 − 2.5 = −0.54) ≈ 0.71 (z-table).

What increases power?

1. **A bigger effect.** +200 ml is easier to detect than +20 ml.
2. **A bigger sample.** SE = σ / √n shrinks, so the center of z (effect / SE) moves away from 0.
3. **Less noise** (smaller σ): more precise measurement, or a paired design like ours.
4. **A larger α.** It is easier to reject H_0, but there are more false alarms.



n	SE, ml	center of z	power
25	80	1.25	0.24
100	40	2.5	0.71
126	36	2.8	0.80
400	20	5	≈ 1.00

Coffee example: real effect +100 ml, σ = 400 ml, α = 0.05. Researchers usually plan for power **0.8**: here that needs about **126 students**.

α and β pull in opposite directions

Coffee example, n = 100, real effect +100 ml:

α	critical value	P(Type I error)	power	P(Type II error)
0.05	1.96	5%	0.71	29%
0.01	2.58	1%	0.47	53%

- A stricter α means fewer false alarms but more missed effects. For a fixed sample you cannot make both errors small at once.
- The only way to reduce both: collect more data, or measure more precisely.
- **"Fail to reject" ≠ "no effect".** With n = 25 the power is only 0.24: in 3 of 4 such studies a real effect stays hidden. A non-significant result from a small study tells us very little.
- **Statistically significant ≠ important.** With a huge n even a tiny effect is detected. Always report the size of the effect.

Learning check

1. A study finds no significant effect ($p = 0.30$). What can we conclude?

a. The effect is exactly zero.

b. There is not enough evidence; a real effect could have been missed.

c. A Type I error happened.

d. H_0 is true with probability 0.30.

2. A researcher changes α from 0.05 to 0.01 (same sample size). What happens?

a. Fewer false alarms and higher power.

b. Fewer false alarms and lower power.

c. More false alarms and higher power.

d. Nothing changes.

3. Which change increases the power of a test?

a. A smaller sample.

b. A larger sample.

c. A smaller α.

d. A noisier measurement.

4. You test 40 questions where in reality nothing is going on, each with $\alpha = 0.05$. How many "significant" results do you expect?

a. 0

b. 1

c. 2

d. 40

Chi-Square Goodness of Fit

Goodness-of-fit test: the idea

One categorical variable: we compare how many individuals fall into each category with what a hypothesis predicts. For example:

- Are there as many women as men in a profession (50/50)?
- Are births spread evenly over the 7 days of the week?
- Does birth weight follow a normal distribution?

H_0: the population has the specified distribution. **H_1:** it does not.

Observed frequencies f_o: the counts in our sample. **Expected frequencies $f_e = n \cdot p$:** the counts we would expect if H_0 were exactly true (p = share of the category under H_0).

$$\chi^2 = \Sigma \frac{(f_o - f_e)^2}{f_e}$$

Small differences between f_o and f_e: random error. Large differences: H_0 does not fit the data.

The Chi-Square Distribution

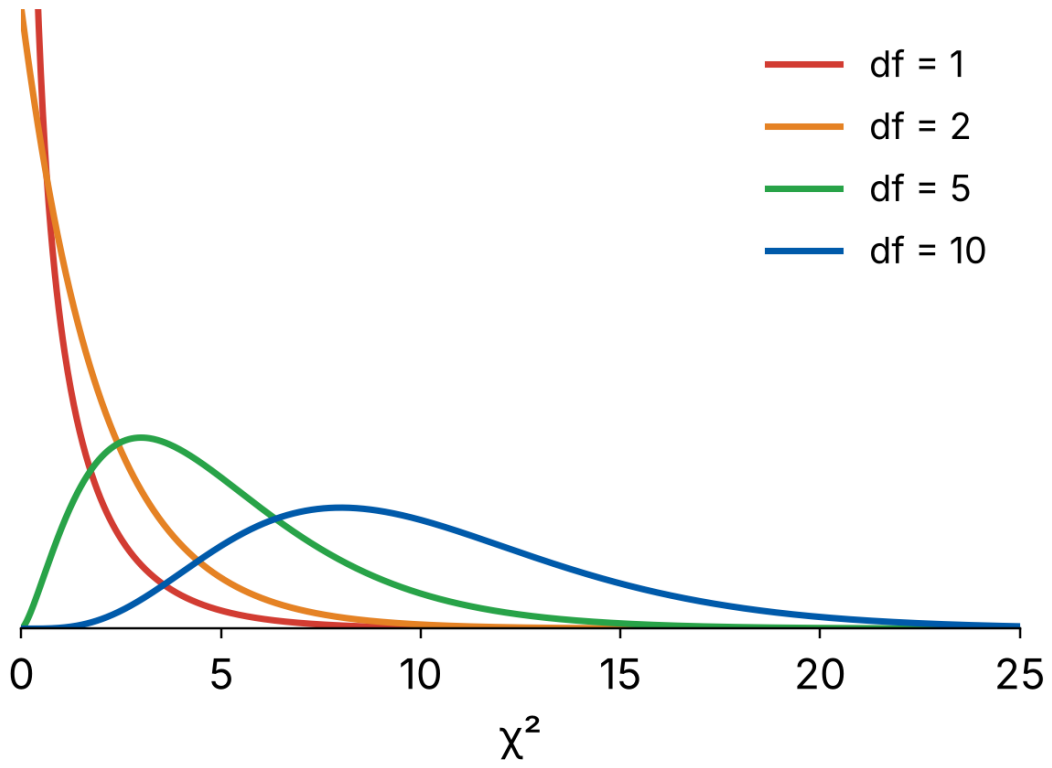
What is the chi-square distribution?

Take k independent z-scores Z_1, …, Z_k (standard normal: mean 0, standard deviation 1). Square them and add them up:

$$\chi^2 = Z_1^2 + Z_2^2 + \ldots + Z_k^2$$

The distribution of this sum is the **chi-square distribution with k degrees of freedom** (df = k). Here df is simply the number of independent squared z-scores in the sum.

- It is never negative: it is a sum of squares.
- **Its mean equals df.** Z has mean 0, so the average of Z^2 is its variance, which is 1. k terms give k on average.
- More terms → the curve moves to the right and becomes more bell-shaped.

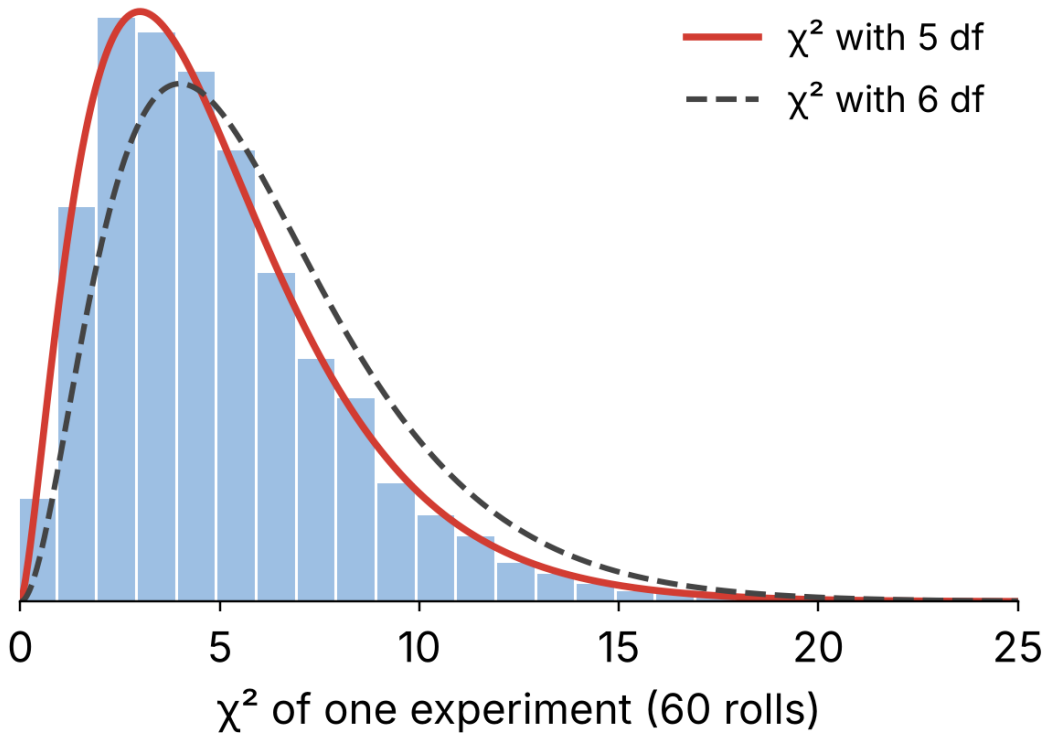


Why does our test statistic follow it?

In each category the count f_o fluctuates around f_e by about $\sqrt{f_e}$. So $(f_o - f_e) / \sqrt{f_e}$ is roughly a z-score, and $\chi^2 = \Sigma\, (f_o - f_e)^2 / f_e$ is a sum of k squared z-scores.

But the k counts are **not free**: they must add up to n. Once k − 1 counts are known, the last one is fixed and brings no new randomness.

So χ^2 behaves like a sum of only **k − 1** independent squared z-scores: **df = k − 1** (Karl Pearson, 1900).



Degrees of freedom: the idea

Degrees of freedom (df) = how many numbers can still change freely after the constraints are fixed.

Example 1. Three friends split a bill of 3 000 ₽. The first can pay any amount, say 1 200 ₽; the second any amount, say 1 000 ₽. The third has no choice: 800 ₽.

Friend 1	Friend 2	Friend 3	Total
1 200 (free)	1 000 (free)	800 (fixed)	3 000

3 numbers − 1 constraint = **2 df**

Example 2. Five numbers must have mean 10, so their sum is 50. Choose four of them freely: 8, 12, 7, 11. The fifth must be 50 − 38 = 12.

1st	2nd	3rd	4th	5th
8	12	7	11	12 (fixed)

5 numbers − 1 constraint = **4 df**

Rule: df = number of values − number of constraints. A constraint is anything that ties the values together: a known total, a known mean, a number computed from the same data.

Why df matters

χ^2 is a sum of squared z-scores, and each free one adds about 1 on average. So **if H_0 is true, $\chi^2 \approx$ df on average**.

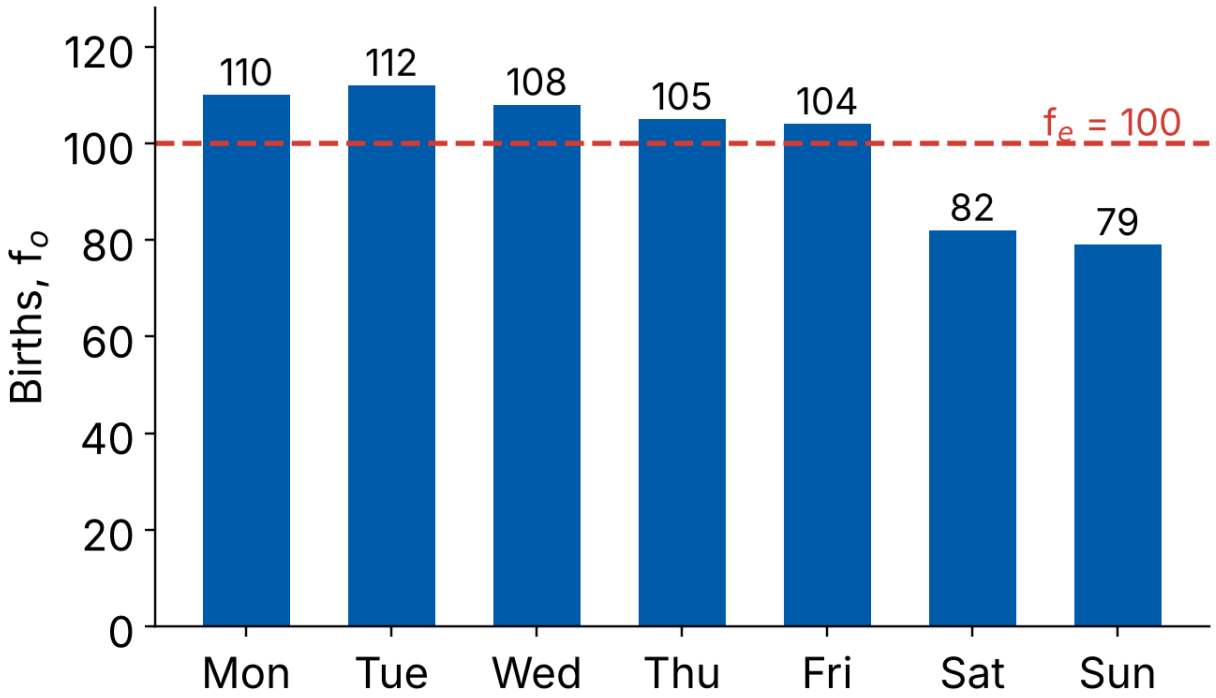
More free categories → more terms → a larger χ^2 by pure chance. That is why the critical value grows with df (table).

Rule of thumb: χ^2 close to df is ordinary; χ^2 several times larger than df is suspicious.

df	α = 0.05	α = 0.01
1	3.84	6.63
2	5.99	9.21
3	7.81	11.34
4	9.49	13.28
5	11.07	15.09
6	12.59	16.81
7	14.07	18.48
8	15.51	20.09
9	16.92	21.67
10	18.31	23.21

Critical values of χ^2 (right tail).

Example 1: births by day of the week



Illustrative data.

A maternity hospital registered n = 700 births. Is every day of the week equally likely?

H_0: every day has the same share p = 1/7. **H_1:** the shares differ.

$f_e = 700 \times 1/7 = 100$ births for every day.

Fewer births on Saturday and Sunday? This is plausible: planned caesarean sections and inductions are rarely scheduled at weekends. But is the dip larger than random error?

Calculating χ^2 for births

Day	f_o	f_e	$(f_o - f_e)^2 / f_e$
Mon	110	100	1.00
Tue	112	100	1.44
Wed	108	100	0.64
Thu	105	100	0.25
Fri	104	100	0.16
Sat	82	100	3.24
Sun	79	100	4.41
Total	**700**	**700**	**$\chi^2 = 11.14$**

df = k − 1 = 7 − 1 = 6: the counts must add up to 700, nothing is estimated. Critical value (α = 0.05): 12.59.

$\chi^2 = 11.14 < 12.59 \rightarrow$ **fail to reject H_0** ($p \approx 0.08$).

Same pattern with **twice the births** (n = 1 400): every count and every term doubles → $\chi^2 = 22.28 > 12.59 \rightarrow$ **reject H_0** ($p \approx 0.001$).

The weekend dip did not change, only the sample size did. With n = 700 the test simply had too little **power**.

Example 2: is birth weight normal?

We have the birth weights of n = 200 newborns. Sample mean M = 3 400 g, sample standard deviation s = 500 g.

Many methods assume a normal distribution, and the shape matters in public health: very light newborns are a risk group.

H_0: birth weight in the population is normally distributed.

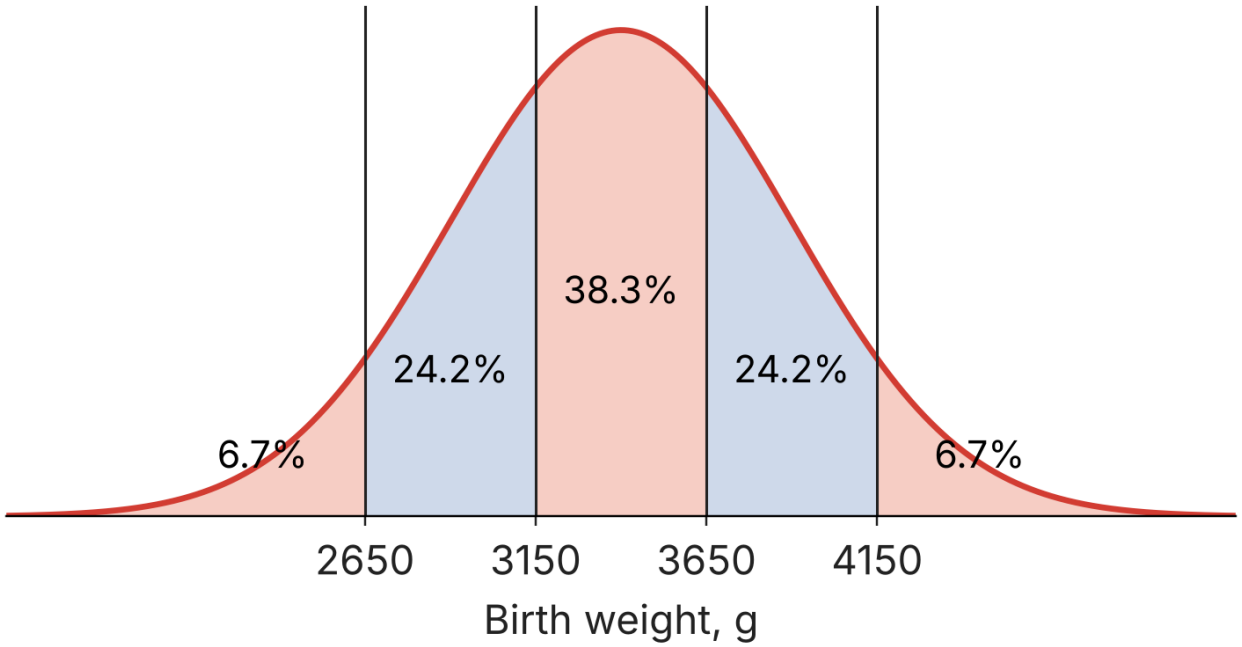
H_1: it is not normally distributed.

Step 1. Split the weights into k = 5 categories and count the newborns in each category: these are the **observed frequencies f_o**.

The boundaries are placed 0.5 and 1.5 standard deviations from the mean: 3 400 ± 250 g and 3 400 ± 750 g.

Birth weight, g	**Newborns, f_o**
< 2 650	19
2 650 – 3 150	32
3 150 – 3 650	81
3 650 – 4 150	59
≥ 4 150	9
Total	**200**

Step 2: shares under the normal curve



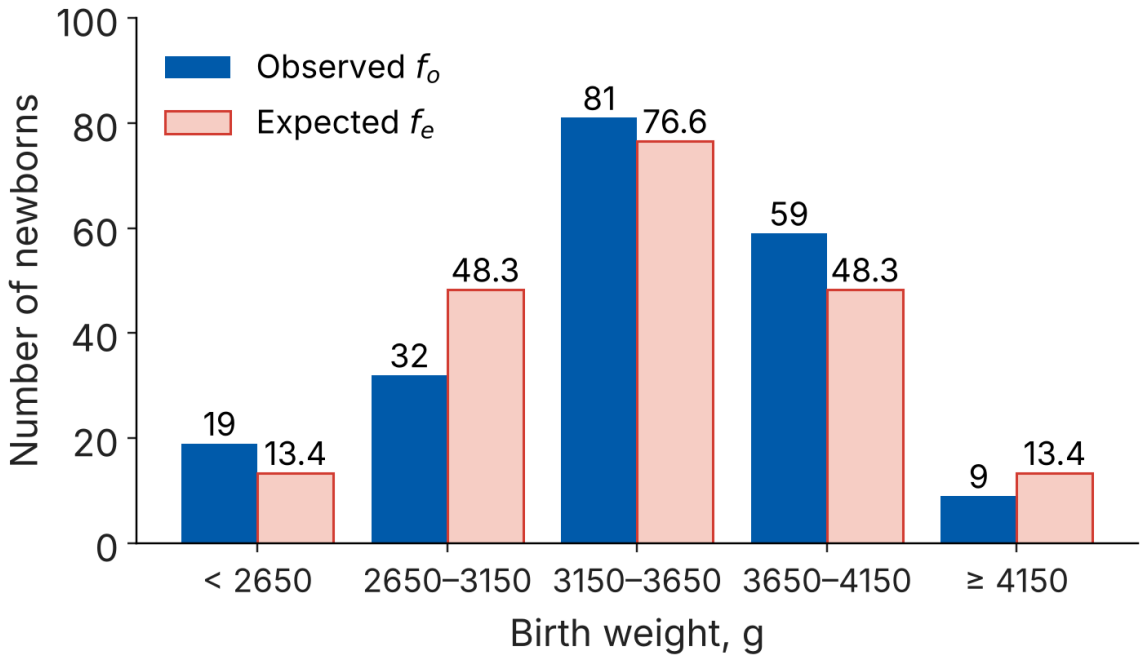
Boundary x, g	z = (x − M) / s	Share of the curve to the left
2 650	−1.5	6.7%
3 150	−0.5	30.9%
3 650	+0.5	69.1%
4 150	+1.5	93.3%

If H_0 is true, the weights follow a normal curve with mean 3 400 g and standard deviation 500 g. We turn each boundary into a **z-score**: its distance from the mean in standard deviations.

A z-table (or software) gives the share of the normal curve to the left of each z. The share of a category is the difference between its two boundaries, for example 30.9% − 6.7% = 24.2%.

Step 3: expected frequencies

Birth weight, g	Share p	$f_e = n \cdot p$
< 2 650	0.0668	200 × 0.0668 = 13.4
2 650 – 3 150	0.2417	200 × 0.2417 = 48.3
3 150 – 3 650	0.3829	200 × 0.3829 = 76.6
3 650 – 4 150	0.2417	200 × 0.2417 = 48.3
≥ 4 150	0.0668	200 × 0.0668 = 13.4
Total	**1**	**200**



f_e is the number of newborns we would expect in each category **if birth weight were exactly normal**. Expected frequencies do not have to be whole numbers.

The plot compares the data (f_o, blue) with the hypothesis (f_e, pink). Are these differences just random error, or too big for that?

Step 4: χ^2, df and decision

Birth weight, g	f_o	f_e	$f_o - f_e$	$(f_o - f_e)^2$	$(f_o - f_e)^2 / f_e$
< 2 650	19	13.4	5.6	31.36	2.34
2 650 – 3 150	32	48.3	−16.3	265.69	5.50
3 150 – 3 650	81	76.6	4.4	19.36	0.25
3 650 – 4 150	59	48.3	10.7	114.49	2.37
≥ 4 150	9	13.4	−4.4	19.36	1.44
Total	**200**	**200**	**0**		**$\chi^2 \approx 11.9$**

- **df = k − 1 − 2 = 5 − 1 − 2 = 2.** The counts add up to n (1 constraint), and the normal curve uses M and s computed from the same data (2 more constraints).
- Critical value (α = 0.05, df = 2): 5.99. $\chi^2 \approx 11.9 > 5.99$ → **reject H_0** ($p \approx 0.003$).
- Birth weight is **not** normal: too few newborns in 2 650 – 3 150 g, too many very light ones (< 2 650 g) and too many in 3 650 – 4 150 g. The distribution is asymmetric, with a long tail of very light babies.

Chi-Square Test for Independence

Two categorical variables

Each of n = 200 students is classified **twice**: by personality type (introvert or extrovert) and by favourite colour. **Question:** is colour preference related to personality?

	Red	**Yellow**	**Green**	**Blue**	**Total**
Introverts	10	3	15	22	50
Extroverts	90	17	25	18	150
Total	**100**	**20**	**40**	**40**	**200**

The numbers inside the table are the observed frequencies f_o. The row and column totals are called the **margins**.

The goodness-of-fit test looked at **one** variable. Now we have **two** variables and ask whether they are independent.

Example: Gravetter & Wallnau, Statistics for the Behavioral Sciences

The null hypothesis for independence

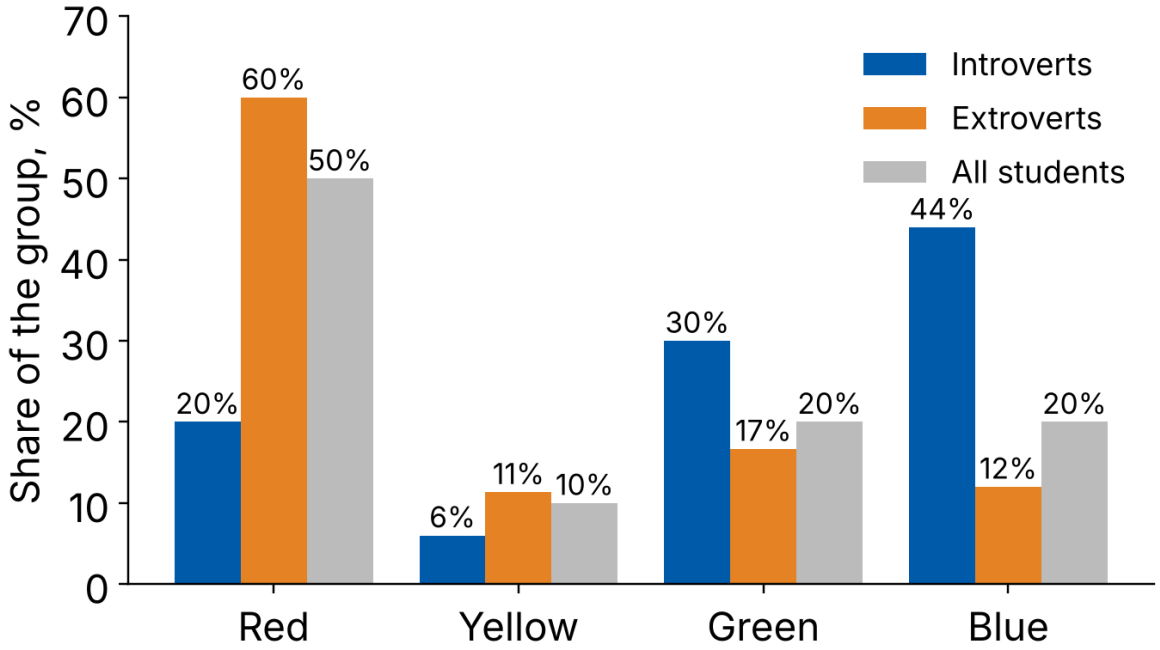
H_0: in the population there is no relationship between colour preference and personality.

In other words: colour preferences have the **same proportions** for introverts and for extroverts.

H_1: the proportions differ: colour preference depends on personality.

When are two variables independent? When knowing one tells you nothing about the other: the distribution of colours looks the same in every row.

In our sample: 44% of introverts but only 12% of extroverts prefer blue. Is this difference too big to be random error?



Expected frequencies

If H_0 is true, introverts and extroverts prefer colours in the **same proportions as all students together**.

1. Shares among all 200 students: red 100 / 200 = 50%, yellow 10%, green 20%, blue 20%.
2. Apply these shares to each row. Introverts: 50% of 50 = 25, 10% of 50 = 5, then 10 and 10. Extroverts: 50% of 150 = 75, then 15, 30 and 30.
3. In one formula (share of the column × row total), for example introverts × red: 50 × 100 / 200 = 25.

$$f_e = \frac{\text{row total} \times \text{column total}}{n}$$

f_e	Red	Yellow	Green	Blue	Total
Introverts	25	5	10	10	50
Extroverts	75	15	30	30	150
Total	**100**	**20**	**40**	**40**	**200**

Calculating χ^2 for the table

Cell	f_o	f_e	$(f_o - f_e)^2 / f_e$
Introverts × red	10	25	9.00
Introverts × yellow	3	5	0.80
Introverts × green	15	10	2.50
Introverts × blue	22	10	14.40
Extroverts × red	90	75	3.00
Extroverts × yellow	17	15	0.27
Extroverts × green	25	30	0.83
Extroverts × blue	18	30	4.80
Total	**200**	**200**	**$\chi^2 = 35.6$**

The same formula as in the goodness-of-fit test, summed over all 8 cells:

$$\chi^2 = \Sigma\, (f_o - f_e)^2 / f_e$$

The biggest terms: introverts × blue (22 instead of 10) and introverts × red (10 instead of 25).

Degrees of freedom for a table

	Red	Yellow	Green	Blue	Total
Introverts	10	3	15	22	50
Extroverts	90	17	25	18	150
Total	**100**	**20**	**40**	**40**	**200**

White cells: free. Grey cells: fixed by the totals.

The totals are fixed. How many cells can we fill in freely?

Fill 3 cells of the introvert row. The 4th is fixed by the row total: 50 − 10 − 3 − 15 = 22.

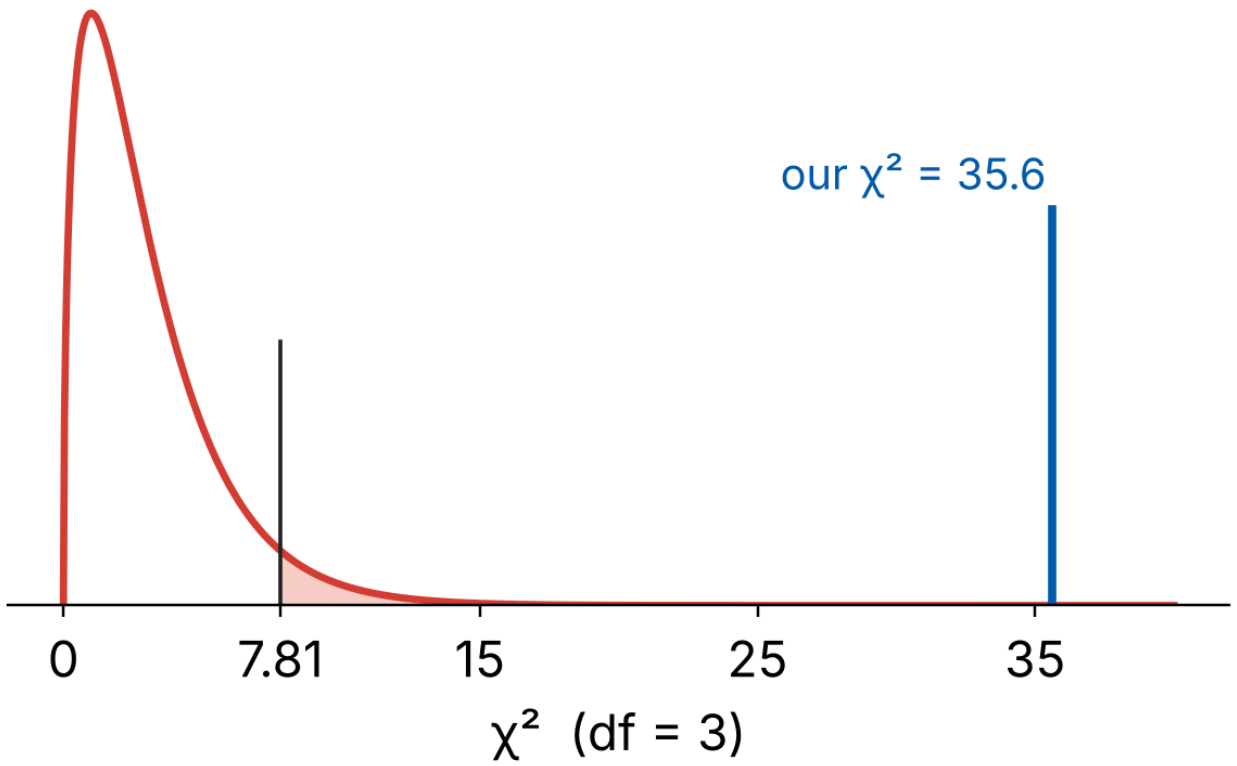
Every extrovert cell is fixed by its column total: 100 − 10 = 90, 20 − 3 = 17, and so on.

Only 3 cells are free: **df = 3**.

In general: in each row the last cell is fixed by the row total (C − 1 free cells per row), and the last row is fixed by the column totals (R − 1 free rows).

$$df = (R - 1)(C - 1) = (2 - 1)(4 - 1) = 3$$

Decision and conclusion



df = 3, α = 0.05 → critical value 7.81.

χ^2 = 35.6 > 7.81 → **reject H_0** (p < 0.001).

Conclusion: colour preference and personality are **related**. Introverts choose blue more often and red less often than extroverts.

Caution: the test shows **that** there is a relationship, not **why**, and not how strong it is. A relationship is not a proof of cause and effect.

Learning check

1. A table has 3 rows and 4 columns. How many degrees of freedom does the test for independence have?

a. 12

b. 11

c. 7

d. 6

2. Goodness of fit with 5 categories; the hypothesis gives all the shares, nothing is estimated. df = ?

a. 5

b. 4

c. 3

d. 2

3. If H_0 is true, a χ^2 statistic with df = 6 is on average about …

a. 0

b. 1

c. 6

d. 36

4. In a test for independence the row total is 40, the column total is 30 and n = 200. The expected frequency of this cell is …

a. 6

b. 12

c. 15

d. 70

Thank you!